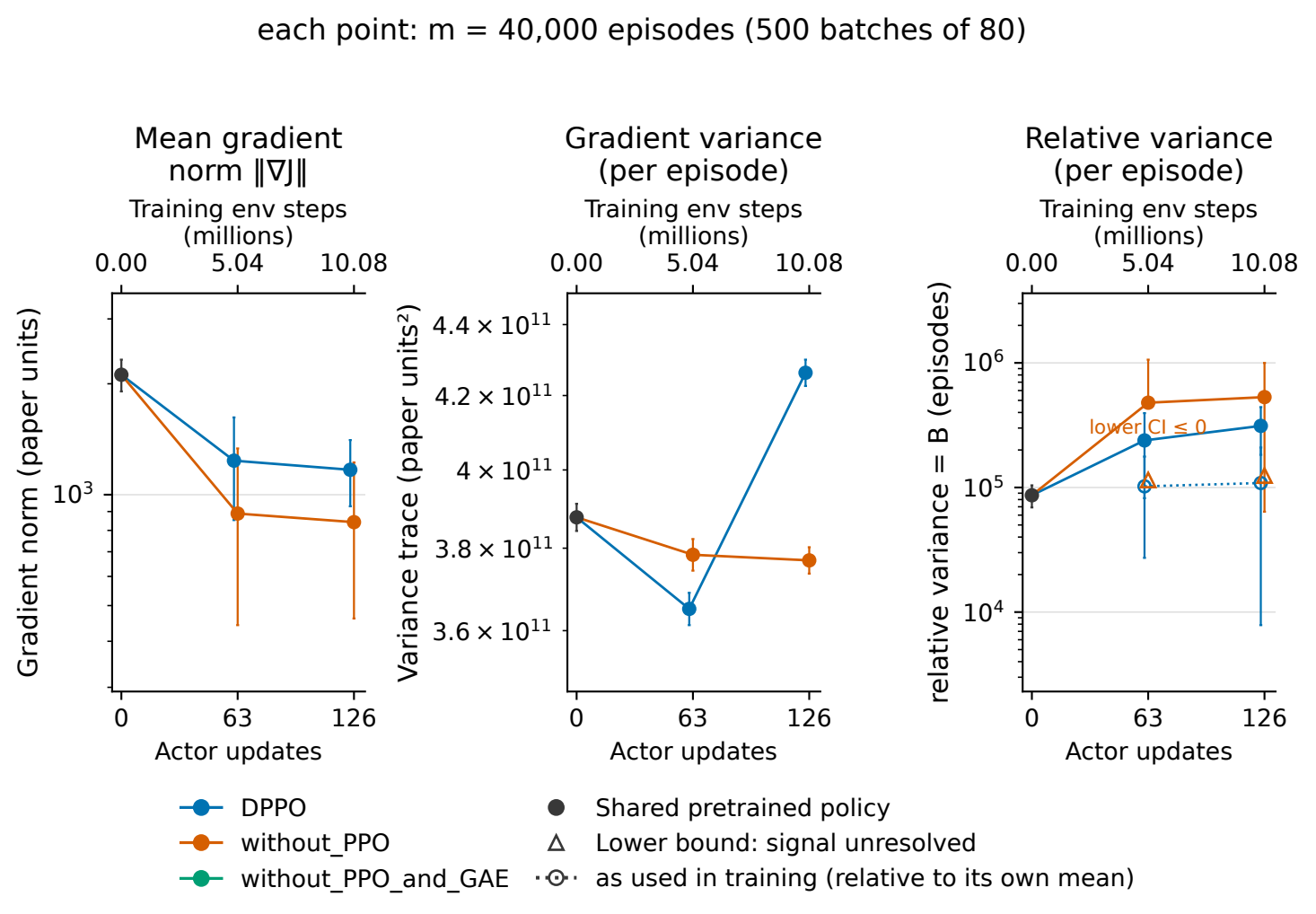
each point: m = 40,000 episodes (500 batches of 80)
Mean gradient norm ‖∇J‖
Training env steps (millions)
0.00
5.04
10.08
Gradient norm (paper units)
10^3
0
63
126
Actor updates
Gradient variance (per episode)
Training env steps (millions)
0.00
5.04
10.08
Variance trace (paper units²)
4.4×10^{11}
4.2×10^{11}
4×10^{11}
3.8×10^{11}
3.6×10^{11}
0
63
126
Actor updates
Relative variance (per episode)
Training env steps (millions)
0.00
5.04
10.08
relative variance = B (episodes)
10^6
10^5
10^4
lower CI ≤ 0
0
63
126
Actor updates
DPPO
without_PPO
without_PPO_and_GAE
Shared pretrained policy
Lower bound: signal unresolved
as used in training (relative to its own mean)